

First Terminal Examination - 2079

Class : XII

Subject: Chemistry (3021)

Full Marks: 75

Time: 3 hrs.

SET 'B'

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Circle the best alternatives.

[11 × 1 = 11]

1. In the reaction, $\text{H}_2 + \text{I}_2 \rightarrow 2\text{HI}$, the rate of disappearance of I_2 is found to be $10^{-6} \text{ mol L}^{-1}\text{s}^{-1}$. The rate of appearance of HI is

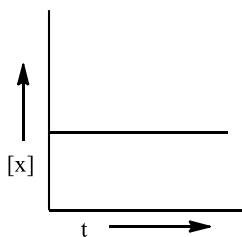
a. $10^{-6} \text{ mol L}^{-1}\text{s}^{-1}$

c. $3 \times 10^{-6} \text{ mol L}^{-1}\text{s}^{-1}$

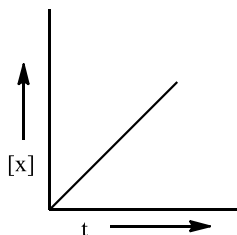
b. $2 \times 10^{-6} \text{ mol L}^{-1}\text{s}^{-1}$

d. $4 \times 10^{-6} \text{ mol L}^{-1}\text{s}^{-1}$
2. Which curve represent zero order reaction

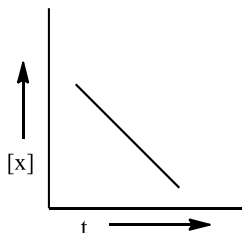
a.



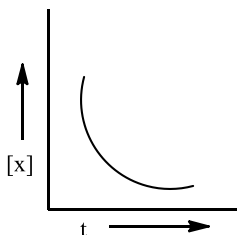
b.



c.



d.



- What is the concentration of the reactant in a first order reaction when the rate of reaction is 0.6 s^{-1} and the rate of constant is 0.035?
 - 26.667 M
 - 17.143 M
 - 26.183 M
 - 17.667 M
- In which of the following conversions, phosphorus pentachloride is used as the reagent?
 - $\text{H}_2\text{C} = \text{CH}_2 \rightarrow \text{CH}_3\text{CH}_2\text{Cl}$
 - $\text{CH}_3\text{CH}_2\text{OH} \rightarrow \text{CH}_3\text{CH}_2\text{Cl}$
 - $\text{H}_3\text{C} - \text{O} - \text{CH}_3 \rightarrow \text{CH}_3\text{Cl}$
 - $\text{CH} \equiv \text{CH} \rightarrow \text{CH}_2 = \text{CHCl}$
- The reaction of tert butyl bromide with sodium methoxide produces mainly
 - isobutane
 - isobutylene
 - tert-butylmethyl ether
 - sodium tert butoxide
- Percentage of carbon in cast iron?
 - 0.1 to 0.25%
 - 2.5 to 5%
 - 0.25 to 2%
 - 1.5 to 3%
- Which of the following compounds can yield only one monochlorinated product upon free radical chlorination?
 - 2, 2-dimethylpropane
 - 2-methylpropane
 - 2-methylbutane
 - n-butane
- Most reactive alkyl halide towards $\text{S}_\text{N}1$ reaction is,
 - n-butyl chloride
 - sec-butyl chloride
 - tert-butylchloride
 - neo-pentylchloride
- In which of the following compounds, Markovnikov's rule does not apply when HCl is added?
 - Propene
 - But-1-ene
 - Pent-2-ene
 - But-2-ene

10. Which of the following has highest boiling point?
- a. 2-chloro-2-methylpropane b. 1-chlorobutane
c. 1-chloro-2-methylpropane d. 2-chlorobutane
11. Which of the following is an example of an ambident nucleophile?
- a. OH^- b. NO_2^-
c. SH^- d. RCO^-

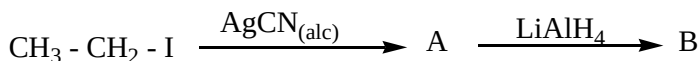
Group - B

Give short answer to the following questions.

[8 × 5 = 40]

1. a. Explain how temperature and catalyst affect the rate of a chemical reaction. [2]
b. Derive integrated rate equation for zero order reaction. Show that half life of zero order reaction is directly proportional to initial concentration of reactant. [2+1]
2. a. Define first order reaction with an example. A first order reaction will takes 100 minutes to complete 60% of reactant into product. What time will it take to complete 90% of reactant into product? [1+2]
b. What are bimolecular reactions? For the second order reaction $\text{P} + \text{Q} \rightarrow \text{Product}$; write two possible rate law equations. [2]
3. a. Explain activation energy and activated complex with the help of energy profile diagram. [3]
b. Calculate the activation energy of a reaction whose rate constant doubles when the temperature raised from 22°C to 32°C. [2]

4. Copper metal is obtained from its ores by using the different metallurgy process. One of its important compound is blue vitriol.
- How is copper matte obtained from roasted ore? Describe with well labelled diagram. [3]
 - How does blue vitriol react with potassium iodide solution? [1]
 - Metallic copper turns green when exposed to moist air. Give reason. [1]
5. The ores containing zinc blend is concentrated and further reduced by vertical retort process to obtain molten zinc.
- How can we obtain pure zinc from zinc spelter? [2]
 - How zinc metal reacts with sodium hydroxide? [1]
 - Write the action of heat on white vitriol. [2]
6. a. Explain the elimination reaction with suitable example. Why chloromethane does not give this reaction? [2+1]
- b. Starting from trichloromethane how would you prepare;
i) chloretone ii) carbonyl chloride [2]
7. How is chlorobenzene prepared from benzene? Why chlorobenzene does not undergo nucleophilic substitution as easily as chloroethane? Mention reasons. Convert chlorobenzene to benzyl amine. [1+3+1]
8. a. How chloroform is prepared from ethanol and bleaching powder paste? [2]
- b. Why chloroform does not give white precipitate with aqueous silver nitrate? [1]
- c. Identify the major product A and B and give IUPAC name. [2]



Group - C

Give long answer to the following questions.

[3 × 8 = 24]

9. a. Chemical kinetics studies the rates and processes of chemical reactions.
- Why are chemists interested in obtaining order of reaction and rate equation? [1]
 - Write differences between rate of reaction and rate constant. [2]
- b. The rate of reaction $2X + Y \rightarrow \text{products}$ is given below as a function of different initial concentration of X and Y.

S.N.	[X] molL ⁻¹	[Y]molL ⁻¹	Initial rate (molL ⁻¹ s ⁻¹)
1	0.10	0.20	3×10^2
2	0.30	0.40	3.6×10^3
3	0.30	0.80	1.44×10^4
4	0.20	0.40	A
5	0.10	0.40	B

- Find out order with respect to X and Y. Also find out overall order.
 - Find out value of rate constant.
 - Find out value of A and B. [3+1+1]
10. Mercury is commonly called quick silver or hydrargyrum. Cinnabar is the only important ore of mercury from which it is extracted
- Explain the working of shaft furnace in the extraction of mercury. [3]
 - How does solid impurities and base metals are removed from impure mercury? [2]
 - What is Nessler's reagent? How is it prepared? Write its use [2]
 - How does calomel react with aqua-regia? [1]

11. a. What is ambident nucleophile? Haloalkanes give alkyl cyanide with KCN but alkyl isocyanides are obtained from AgCN, why? [1+2]
- b. A primary haloalkane 'A' (C_3H_7Cl) when heated with alcoholic KOH produces compound 'B'. 'B' on treatment with HBr produces 'C'. When heated with sodium in ether, 'C' produces Compound 'D'. Identify A, B, C, D and write the reactions involved. How is compound 'C' converted into propan-2-ol? [4+1]

❧ - The End - ❧